

E-COMMERCE SALES PERFORMANCE ANALYSIS USING POWER BI

Project Overview

The objective of this project is to analyze an E-commerce sales dataset using Power BI. The project focuses on transforming raw data into meaningful business insights by creating calculated columns, DAX measures, and interactive dashboards. The dashboard helps stakeholders monitor sales performance, profitability, sales targets, and regional sales trends to support informed business decisions.

Problem Statement

Businesses generate large volumes of sales data every day. Without proper analysis, it becomes difficult to identify profitable products, monitor sales performance, compare actual sales against targets, and understand customer purchasing patterns.

The objective of this project is to build an interactive Power BI dashboard that provides meaningful insights into sales performance using DAX calculations and visual analytics.

Dataset Description

The project uses three datasets.

1. List of Orders

Contains customer and order information.

Attributes

- Order ID
- Order Date
- Customer Name
- City
- State
- Category
- Amount
- Profit

2. Order Details

Contains product-level order details.

Attributes

- Order ID
- Category
- Sub-Category
- Quantity
- Amount
- Profit

3. Sales Target

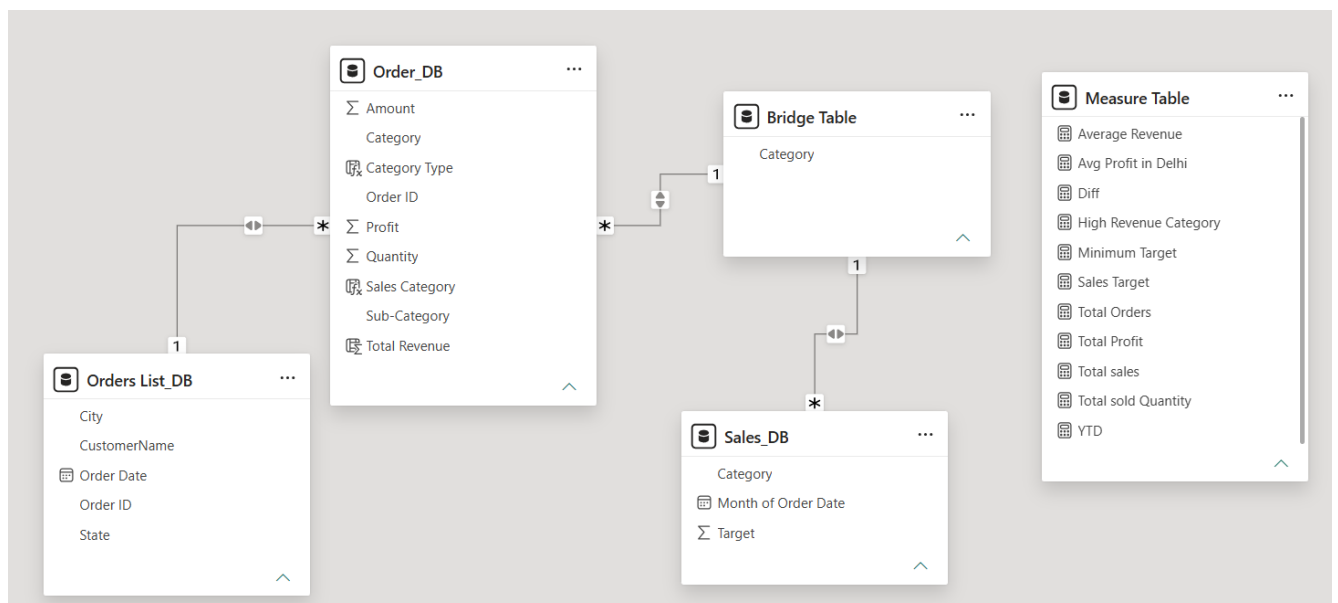
Contains monthly sales targets for each category.

Attributes

- Category
- Month of Order Date
- Target

Data Modeling

The datasets were imported into Power BI and appropriate relationships were established.



The model includes:

- Relationship between Orders and Order list details using **Order ID**
- Bridge table created for **Category** to avoid many-to-many relationships
- Relationship between Bridge Category and Sales Data.

DAX Calculations

Calculated Columns

Category Type

Combined Category and Sub-Category into a single column.

```
Category Type = Order_DB[Category]&" "&Order_DB[Sub-Category]
```

Revenue per Order

Calculated revenue using: Revenue = Amount × Quantity

```
Total Revenue = Order_DB[Quantity]*Order_DB[Amount]
```

Sales Category

Categorized orders into:

- Above Average
- Below Average

based on the average sales amount.

```
Sales Category =  
IF(  
    Order_DB[Total Revenue] >='Measure Table'[Average Revenue],  
    "Above Average",  
    "Below Average"  
)
```

DAX Measures

The following measures were created:

- **Order Count**

```
Total Orders = COUNT(Order_DB[Order ID])
```

- **Total Sales**

```
Total sales = SUM(Order_DB[Total Revenue])
```

- **Total Profit**

```
Total Profit = SUM(Order_DB[Profit])
```

- **Total Quantity**

```
Total sold Quantity = SUM(Order_DB[Quantity])
```

- **Sales Target**

```
Sales Target = SUM(Sales_DB[Target])
```

- **Average Profit in Delhi**

```
Avg Profit in Delhi = CALCULATE(AVERAGE(Order_DB[Profit]),'Orders List_DB'[State]="Delhi")
```

- **Year-to-Date (YTD) Sales**

```
YTD =  
TOTALYTD([Total sales],'Orders List_DB'[Order Date].[Date])
```

- **Minimum Target**

```
Minimum Target = MIN(Sales_DB[Target])
```

These measures were used throughout the dashboard for business analysis.

Dashboard Visualizations

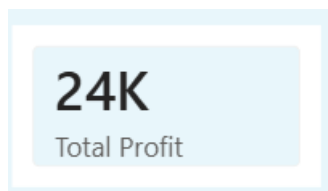
The dashboard contains the following visuals.

KPI Cards

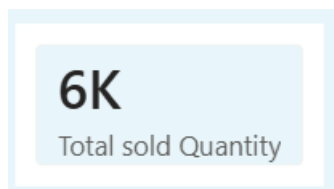
- **Total Sales**



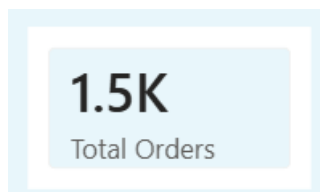
- **Total Profit**



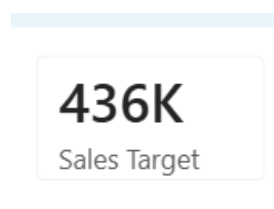
- **Total Quantity**



- **Order Count**

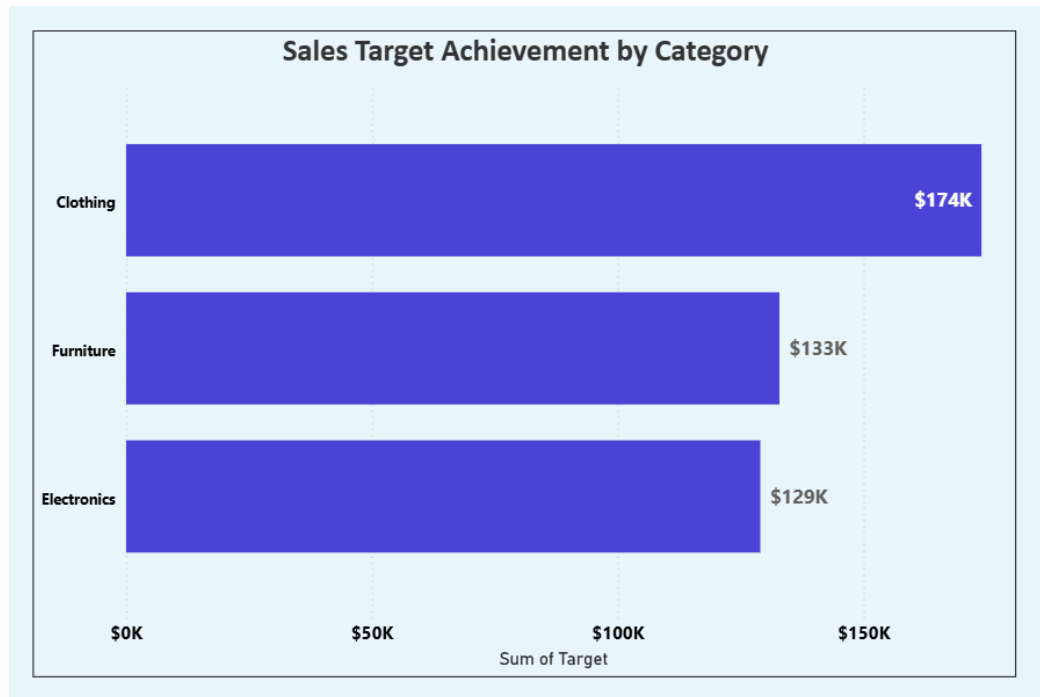


- **Sales Target**



Clustered Column Chart

Sales Target Achievement by Category



This horizontal bar chart shows the **Sales Target Achievement by Category**. It compares the total **sales target** assigned to each product category.

Explanation

- **Clothing** has the **highest sales target**, with a total target of **\$174K**.
- **Furniture** has the **second-highest sales target**, with **\$133K**.
- **Electronics** has the **lowest sales target**, with **\$129K**.

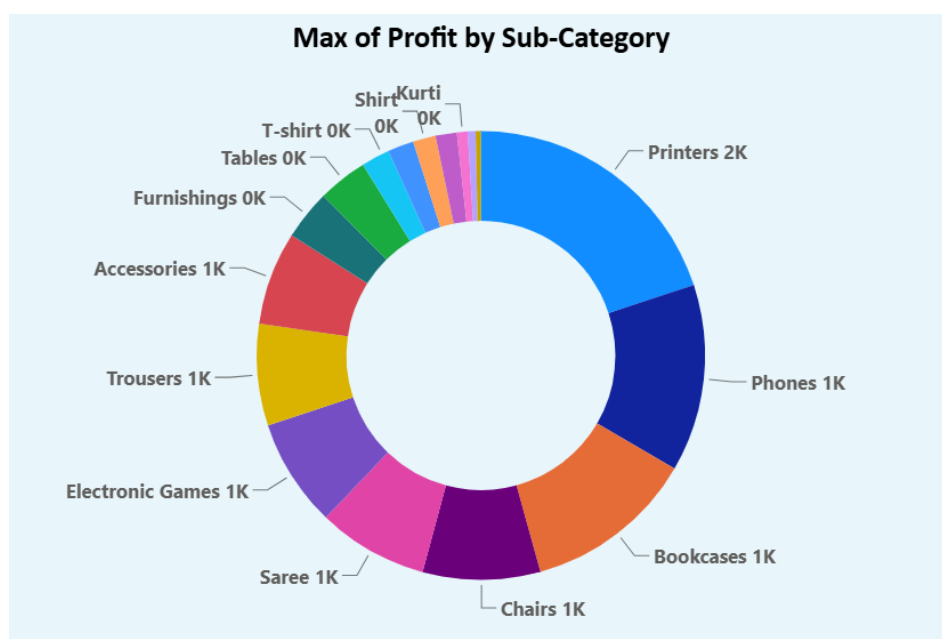
Key Insights

1. **Clothing is the primary focus category**, accounting for the highest target of **\$174K**, indicating the company expects the greatest sales from clothing products.
2. **Furniture and Electronics have similar targets**, with only a **\$4K difference** between them.
3. The **difference between Clothing and Furniture is \$41K**, showing that Clothing has a significantly higher sales expectation.
4. The **difference between Clothing and Electronics is \$45K**, making Electronics the category with the lowest target.
5. Since this chart displays **sales targets only**, it does **not indicate actual sales performance**. To evaluate whether each category met or exceeded expectations, compare these targets with the actual sales values.

Donut Chart

Maximum Profit Margin by Sub-Category

Shows which product categories generate the highest profit margin.



This **doughnut chart** represents the **Maximum Profit earned by each Sub-Category**. It highlights which sub-categories generated the highest single profit value.

Explanation

- **Printers** generated the **highest maximum profit** of approximately **2K**, making it the most profitable sub-category.
- **Phones, Bookcases, Chairs, Saree, Electronic Games, Trousers, and Accessories** each recorded a maximum profit of around **1K**.
- **Furnishings, Tables, T-shirts, Shirts, and Kurtis** contributed **very low maximum profits (less than 1K)**, which appear as **0K** due to rounding in the chart.

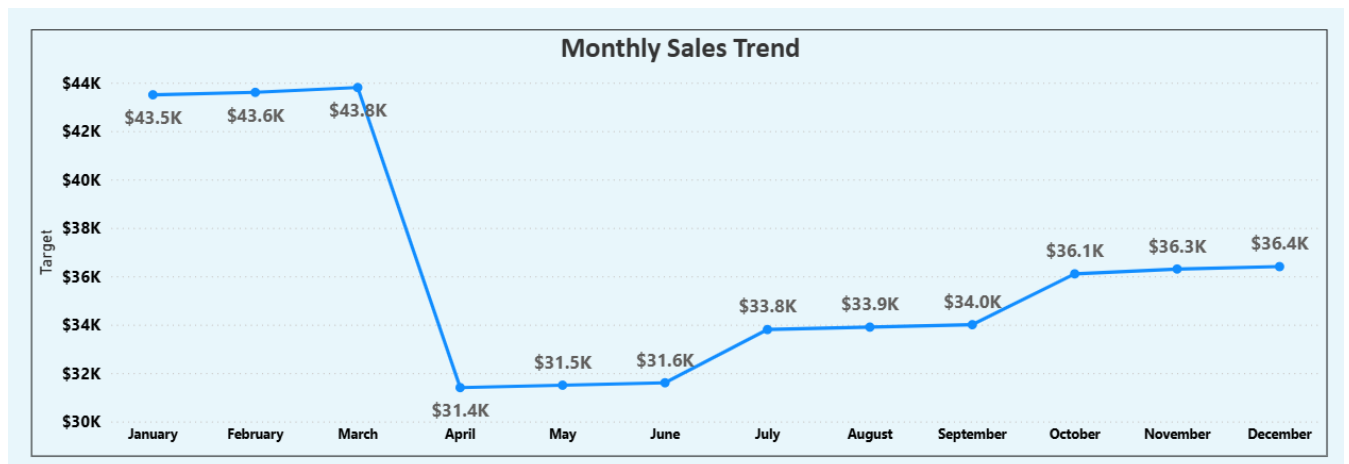
Key Insights

1. **Printers are the top-performing sub-category**, achieving the highest maximum profit (2K).
2. Several sub-categories, including **Phones, Bookcases, Chairs, Saree, Electronic Games, Trousers, and Accessories**, have similar profit performance, each reaching around **1K**.
3. **Furnishings, Tables, T-shirts, Shirts, and Kurtis** contribute the least to maximum profit, indicating lower profitability.
4. The profit distribution is spread across many sub-categories, but **Printers clearly outperform the others**.
5. The chart uses the **maximum profit** value rather than **total profit**, meaning it shows the **highest profit earned from a single transaction or record** for each sub-category, not the overall profit.

Line Chart

Monthly Sales Trend

This **line chart** illustrates the **Monthly Sales Target Trend** throughout the year. It shows how the sales target changes from **January to December**, helping identify seasonal patterns and business planning.



Explanation

- The sales target starts at **\$43.5K** in **January** and gradually increases to **\$43.8K** in **March**, the **highest target** of the year.
- In **April**, the target drops sharply to **\$31.4K**, representing the **lowest target**.
- From **May to September**, the target gradually increases from **\$31.5K** to **\$34.0K**.
- During the final quarter (**October to December**), the target continues to rise, reaching **\$36.4K** in **December**.

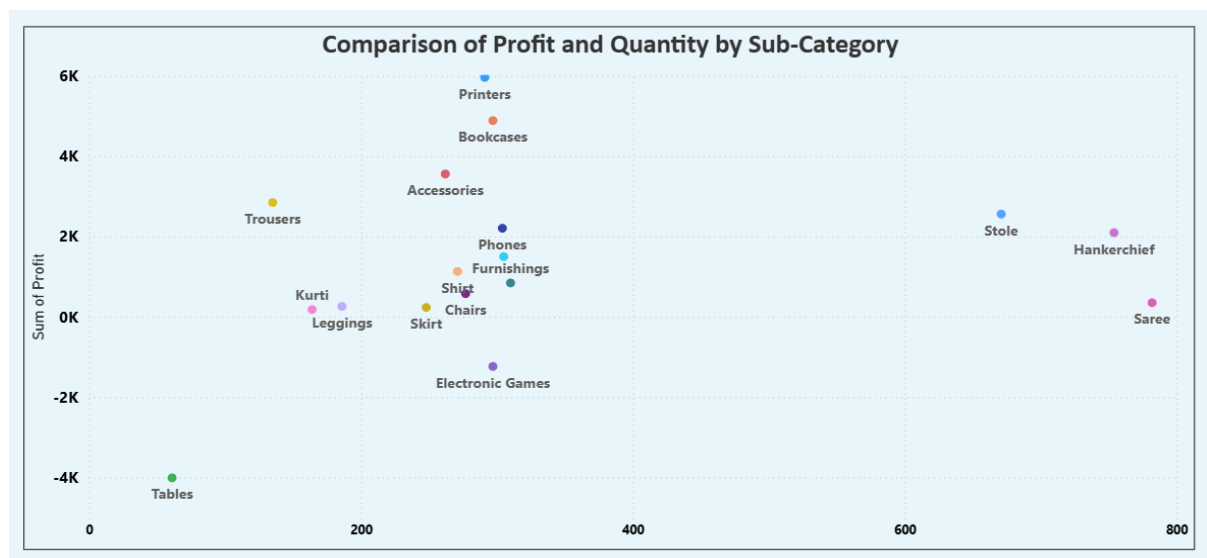
Key Insights

1. **March recorded the highest sales target at \$43.8K**, indicating strong business expectations during the first quarter.
2. **April experienced the largest decline**, with the target falling from **\$43.8K to \$31.4K**—a decrease of approximately **\$12.4K**.

3. After April, the sales target shows a **steady upward trend**, reflecting gradual business recovery or planned growth.
4. **December ends the year with a target of \$36.4K**, which is higher than the mid-year values but still below the first-quarter peak.
5. The trend suggests that the company follows a **seasonal target strategy**, with ambitious targets at the beginning of the year, a mid-year slowdown, and a gradual increase toward year-end.

Scatter Chart

Profit vs Quantity



This **scatter chart** compares the **Total Profit (Y-axis)** and **Quantity Sold (X-axis)** for each product **Sub-Category**. It helps identify which products generate high profits, which sell in large quantities, and which incur losses.

Explanation

- Each point represents a **sub-category**.
- The **X-axis** shows the **Quantity Sold**.

- The **Y-axis** shows the **Total Profit**.
- Points higher on the chart indicate **higher profit**, while points farther to the right indicate **higher sales quantity**.

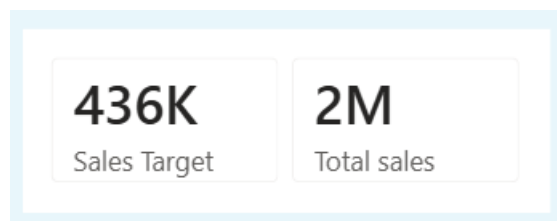
Key Insights

1. **Printers** achieved the **highest profit (around 6K)** while maintaining a moderate sales quantity, making it the **most profitable sub-category**.
2. **Bookcases** generated the **second-highest profit (around 5K)** with a similar sales quantity to Printers.
3. **Stole** and **Handkerchief** recorded **very high sales quantities (around 680–760 units)** while also earning good profits, indicating strong customer demand.
4. **Tables** experienced the **largest loss (around -4K profit)** despite being sold, suggesting poor profitability or high costs.
5. **Electronic Games** also showed a **negative profit (around -1.5K)**, indicating another underperforming product category.
6. **Phones, Accessories, Furnishings, and Shirts** generated **positive profits with moderate sales quantities**, showing balanced performance.
7. **Saree** had one of the **highest sales quantities (around 780 units)** but earned only a **small profit**, indicating that high sales volume does not always result in high profitability.
8. **Kurti, Leggings, Skirt, and Chairs** generated **low profits despite moderate sales quantities**, suggesting opportunities to improve pricing or reduce costs.

Business Interpretation

- **High Profit & Good Sales:** Printers, Bookcases, Phones.
- **High Sales but Low Profit:** Saree, Handkerchief.
- **Loss-Making Products:** Tables and Electronic Games.
- **Average Performers:** Accessories, Furnishings, Shirts, Chairs.

Multi-row Card



The KPI cards compare the **Total Sales (2M)** with the **Sales Target (436K)**, providing a quick overview of business performance. Since actual sales are much higher than the target, the organization has successfully exceeded its sales objective.

Explanation

- **Sales Target: 436K** – This is the planned sales goal set by the organization.
- **Total Sales: 2M** – This represents the actual sales achieved during the selected period.

Key Insight

- The **Total Sales (2M)** significantly **exceed the Sales Target (436K)**, indicating that the organization has **surpassed its sales goal** and achieved outstanding sales performance.

Matrix

Sub-Category	Minimum Target
Hankerchief	₹ 12,000
Kurti	₹ 12,000
Leggings	₹ 12,000
Saree	₹ 12,000
Shirt	₹ 12,000
Skirt	₹ 12,000
Stole	₹ 12,000
Trousers	₹ 12,000
T-shirt	₹ 12,000
Bookcases	₹ 10,400
Chairs	₹ 10,400
Furnishings	₹ 10,400
Tables	₹ 10,400
Accessories	₹ 9,000
Electronic Games	₹ 9,000
Phones	₹ 9,000
Printers	₹ 9,000

This **Multi-Row Card** displays the **Minimum Sales Target** assigned to each **Sub-Category**. It helps identify the lowest target values across different product groups.

Explanation

- **Handkerchief, Kurti, Leggings, Saree, Shirt, Skirt, Stole, Trousers, and T-shirt** have the **highest minimum target** of **₹12,000**.
- **Bookcases, Chairs, Furnishings, and Tables** each have a **minimum target** of **₹10,400**.
- **Accessories, Electronic Games, Phones, and Printers** have the **lowest minimum target** of **₹9,000**.

Key Insights

1. Most clothing-related sub-categories have a **minimum target of ₹12,000**, indicating higher baseline sales expectations.
2. Furniture-related sub-categories have a **moderate minimum target of ₹10,400**.
3. **Accessories, Electronic Games, Phones, and Printers** have the **lowest minimum target (₹9,000)**, suggesting comparatively lower sales expectations.
4. The visual provides a quick reference to compare baseline targets across all product sub-categories.

Matrix

Month	Sales Target
January	43500
February	43600
March	43800
April	31400
May	31500
June	31600
July	33800
August	33900
September	34000
October	36100
November	36300
December	36400
Total	435900

This table displays the **monthly sales targets** and the **overall annual sales target**. It helps track how sales targets are distributed throughout the year.

Explanation

- The **highest monthly sales target** is in **March (43,800)**, followed by **February (43,600)** and **January (43,500)**.
- The **lowest sales target** is in **April (31,400)**.
- From **May to December**, the sales target gradually increases from **31,500** to **36,400**, showing a steady upward trend.
- The **total annual sales target** is **435,900**.

Key Insights

1. **March** has the highest sales target (**43,800**), indicating the strongest sales expectations.
2. **April** records the lowest target (**31,400**), reflecting a significant drop after the first quarter.
3. Sales targets gradually recover from **May onward**, increasing consistently until **December (36,400)**.

4. The **annual sales target** across all months is **435,900**, serving as the benchmark for evaluating yearly sales performance.

Matrix

Category	Sum of Target	Sum of Total Revenue	Diff
Electronics	\$1,29,000	\$8,16,583	687583
Furniture	\$1,32,900	\$6,65,765	532865
Clothing	\$1,74,000	\$6,64,522	490522
Total	\$4,35,900	\$21,46,870	1710970

This **table visual** compares the **Sales Target**, **Actual Total Revenue**, and the **Difference (Revenue – Target)** for each product category. It helps evaluate category-wise sales performance against the planned targets.

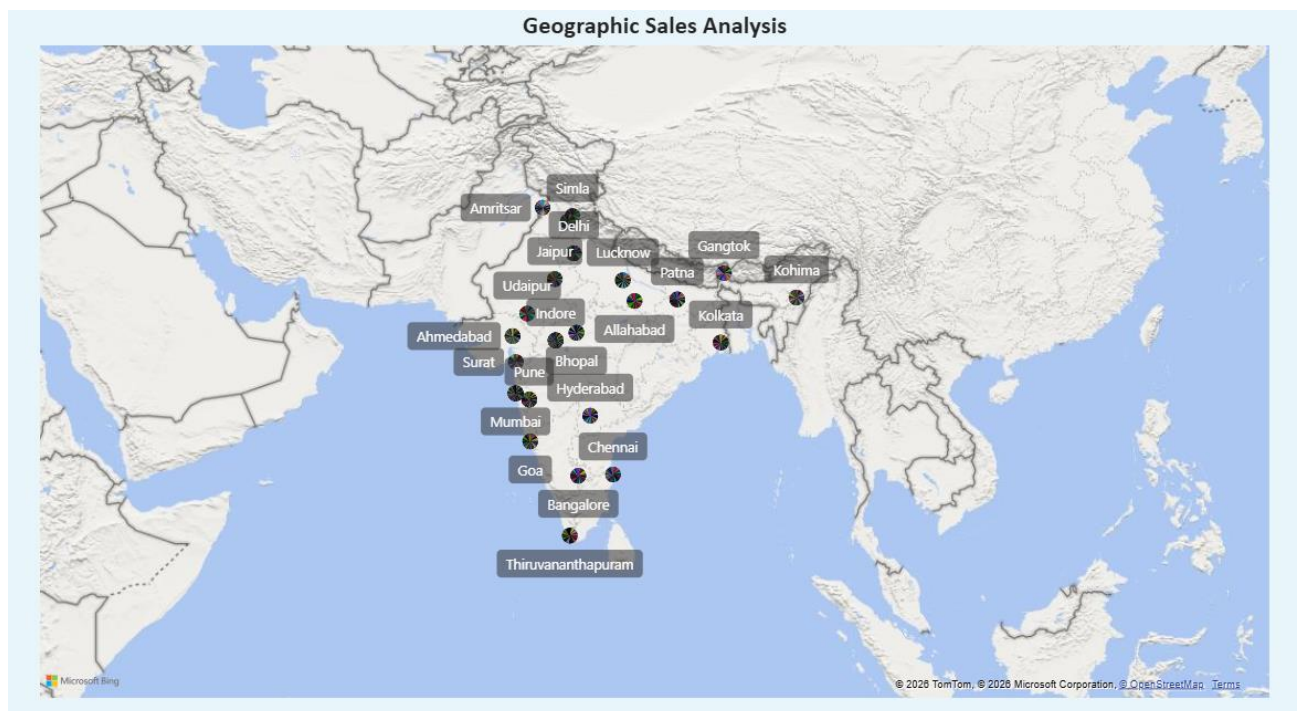
Explanation

- **Electronics** has a sales target of **₹1,29,000** and achieved **₹8,16,583** in revenue, exceeding the target by **₹6,87,583**.
- **Furniture** has a target of **₹1,32,900** and generated **₹6,65,765** in revenue, resulting in a surplus of **₹5,32,865**.
- **Clothing** has the highest sales target of **₹1,74,000** and achieved **₹6,64,522**, exceeding the target by **₹4,90,522**.
- Overall, the **total sales target** is **₹4,35,900**, while the **total revenue** is **₹21,46,870**, producing an overall surplus of **₹17,10,970**.

Key Insights

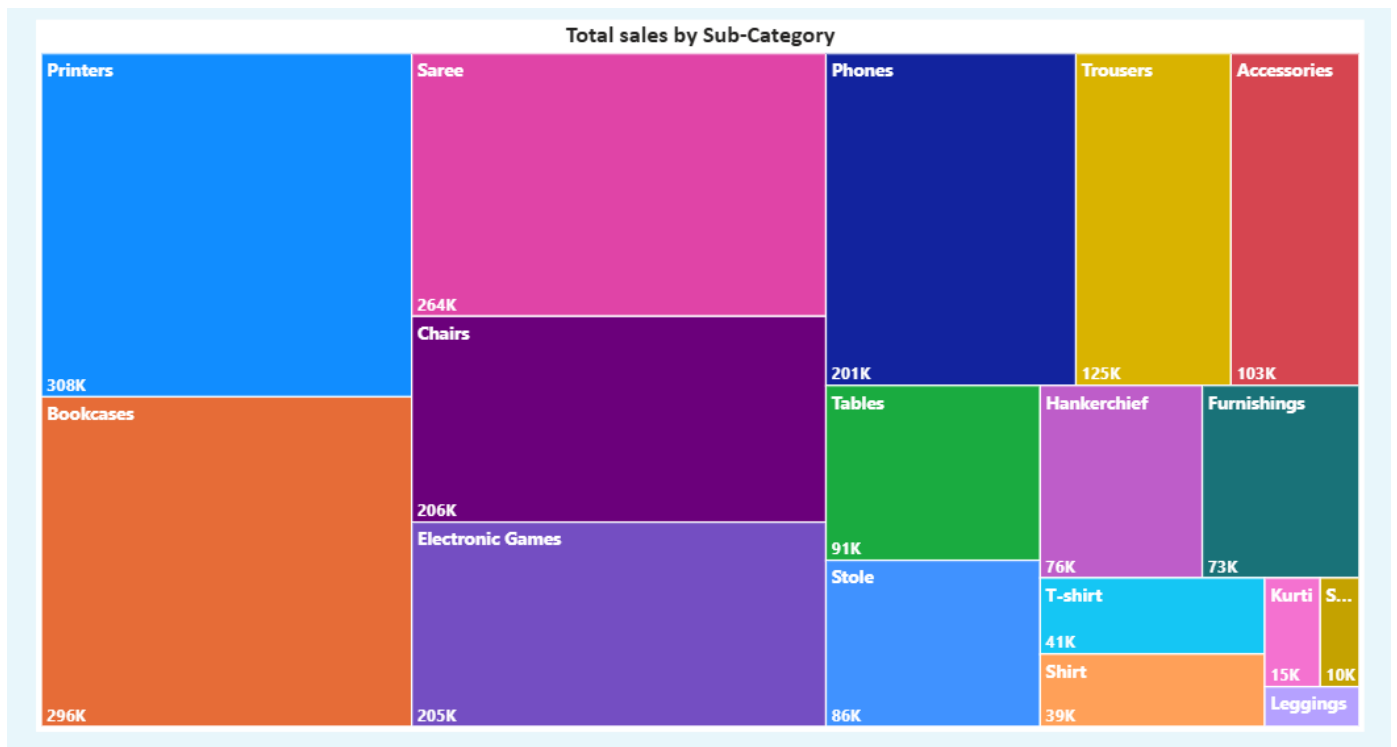
1. **Electronics** generated the highest revenue and recorded the **largest positive difference** from its sales target.
2. **Clothing** had the highest target but generated slightly less revenue than Electronics and Furniture.
3. **All three categories exceeded their sales targets**, indicating strong business performance.
4. The business achieved a **total revenue of ₹21,46,870** against a target of **₹4,35,900**, surpassing the target by **₹17,10,970**.

Map



The map visual displays the **total sales generated across different cities**, providing a geographical view of sales performance. It helps identify cities with the highest and lowest sales contributions. This visualization enables users to analyze regional sales trends and compare performance across locations. It supports data-driven decisions for market expansion and resource allocation.

Tree map



This **Tree map** visual displays the **Total Sales by Sub-Category**, where the size of each rectangle represents the sales contribution of that sub-category.

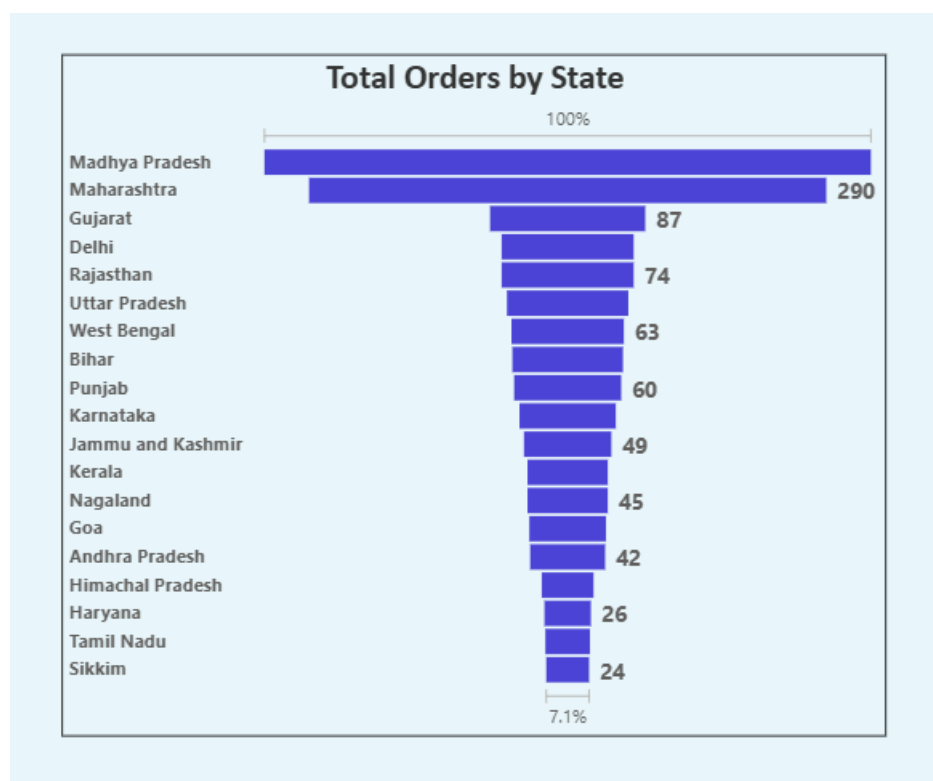
Explanation

- **Printers** recorded the **highest total sales (308K)**, followed by **Bookcases (296K)** and **Saree (264K)**.
- **Phones (201K)**, **Chairs (206K)**, and **Electronic Games (205K)** also contributed significantly to total sales.
- **Kurti (15K)**, **Skirt (10K)**, **Leggings**, and **Shirt (39K)** generated the lowest sales among all sub-categories.
- The treemap provides a quick visual comparison of sales distribution, making it easy to identify top- and low-performing product sub-categories.

Key Insights

1. **Printers** are the highest revenue-generating sub-category.
2. **Bookcases** and **Saree** are also major contributors to overall sales.
3. Sales are concentrated in a few sub-categories, while several products contribute relatively little.
4. The visualization helps identify which products drive the majority of business revenue.

Funnel Chart



This **Funnel Chart** displays the **total number of orders placed across different states**, ranking them from the highest to the lowest order count.

Explanation

- **Madhya Pradesh** recorded the **highest number of orders**, making it the top-performing state.
- **Maharashtra** ranked second with **290 orders**, followed by **Gujarat (87)** and **Rajasthan (74)**.
- States such as **West Bengal (63)**, **Punjab (60)**, and **Jammu and Kashmir (49)** contributed a moderate number of orders.
- **Tamil Nadu (26)** and **Sikkim (24)** recorded the **lowest order volumes**, indicating lower customer activity in these regions.

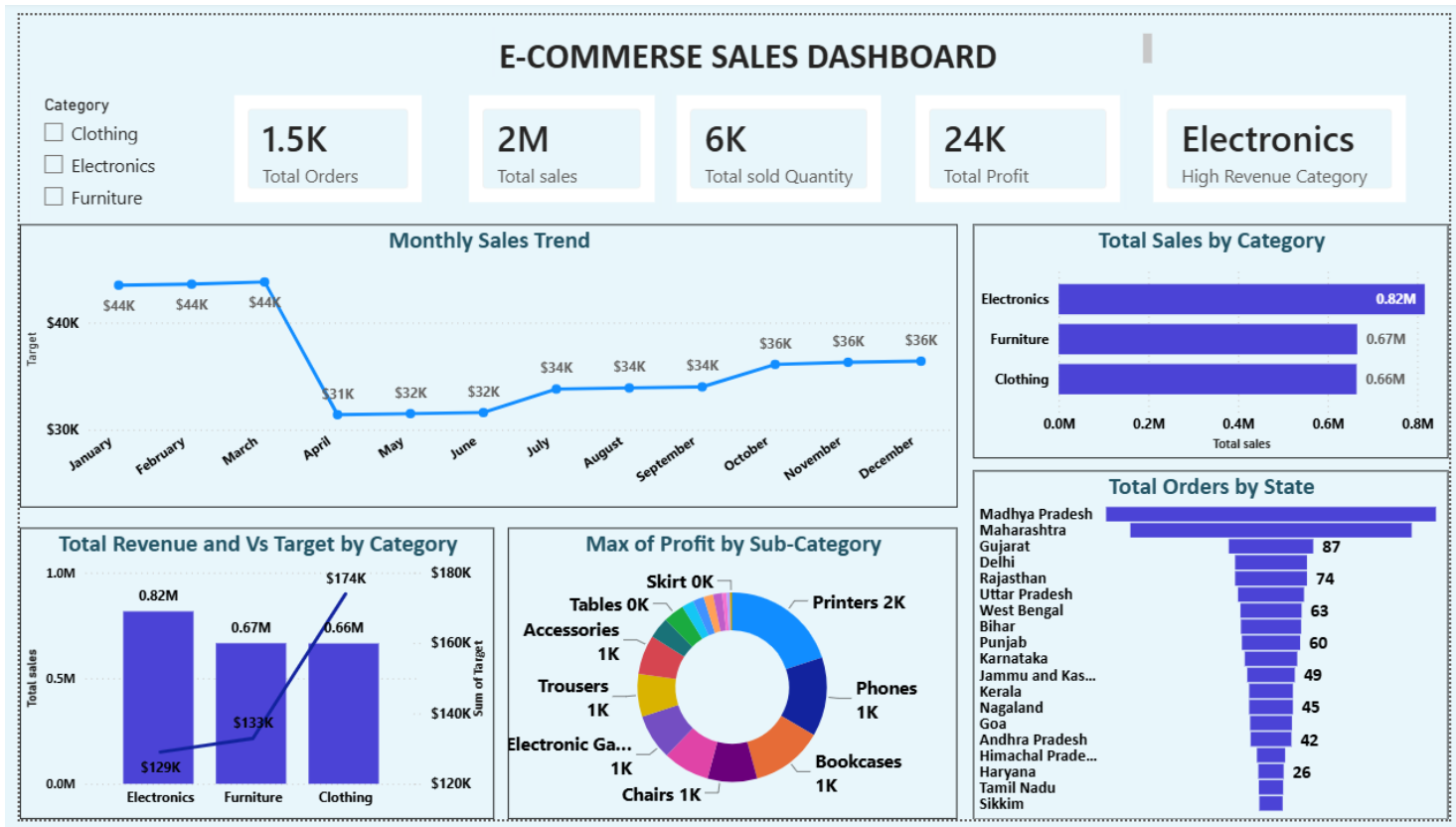
Key Insights

1. **Madhya Pradesh** is the leading state in terms of total orders.
2. **Maharashtra** is the second-highest contributor, while **Gujarat** and **Rajasthan** also show strong order volumes.
3. Order counts decline significantly across the remaining states, indicating an uneven distribution of sales.
4. States with lower order volumes, such as **Tamil Nadu** and **Sikkim**, may offer opportunities for targeted marketing and business expansion.

Tools & Technologies Used

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Modeling
- Microsoft Excel
- Data Visualization

DASHBOARD



Skills Demonstrated

- Data Cleaning
- Data Modeling
- Relationship Management
- Bridge Table Creation
- DAX Calculations
- Time Intelligence (YTD)
- KPI Development
- Dashboard Design
- Business Intelligence
- Data Visualization

Conclusion

This project demonstrates the use of Power BI to transform raw e-commerce data into an interactive business intelligence dashboard. By applying DAX calculations, data modeling, and effective visualizations, the dashboard provides actionable insights into sales performance, profitability, target achievement, and regional trends. It enables stakeholders to make data-driven decisions and monitor key business metrics efficiently.